

3 Methods

A Method is invoked by a method call. The Syntax is as follows:

```
object.method(arguments);
```

Usually, we start with an instance of an object. The object hands the method some information (arguments) and asks that method to perform a task based on that information. When the the task is done, the method returns information back to the object that called it.

```
return = object.method(arguments);
```

Example:

```
Teacher.asksStudent( help );
```

The OBJECT (Teacher) does not need to know how the method (student) accomplished the request. When we're writing software, one object is independent of another. How any object implements its own methods is hidden from the outside world. That way, once your method works, nobody can mess it up. Hiding the implementation of a method promotes good software engineering. If you can't change how an object performs its methods, then you can't introduce errors. You have to live with the way it works now.

A Method can receive as many arguments as you wish. But it can only return ONE thing. A Method also always Receives A Copy Of Its Arguments. Since a Method Only Receives a duplicate, it can NEVER change the original copy of the parameter it received.

```
import java.awt.Container;
import javax.swing.*;
public class SquareInt extends JApplet
{
    public void init()
    {
        String output = "";
        JTextArea oArea = new JTextArea( 10, 20 );
        Container c = getContentPane();
        c.add( oArea );
        int result;
        for( int x = 1; x <= 10; x++ )
        {
            result = square( x );
        }
    }
}
```